5.1.1 Collection of soil and plant samples and analysis

The ICAR-CRIDA team visited the farmers' fields of Visakhapatnam and Vizianagaram districts of Andhra Pradesh, Mandya and Tumkur districts of Karnataka, and Hingoli and Parbhani districts of Maharashtra, interacted with the farmers and collected composite soil samples from the fields of both natural and conventional farming practices after the harvest of crop during April and May, 2019, labeled them properly and brought to ICAR-CRIDA laboratory for air drying, processing and analysis of chemical parameters like organic carbon, available N, P, K, calcium, magnesium, sulphur, available micronutrients (iron, manganese, copper, zinc and boron). Standard protocol for soil analysis is given in Annexure 1. Another set of samples were collected and immediately kept in ice bags for microbial analysis. Again, during the end of August month, ICAR-CRIDA team visited the farmers' fields of Visakhapatnam and Vizianagaram districts of Andhra Pradesh to observe the condition of the standing crops, collected the plant and *jeevamritha* samples, interacted with farmers and other district officials. Likewise, during the first week of December, the team visited farmers' fields of Parbhani and Hingoli districts of Maharashtra to observe the condition of the standing crops, collection of the plant and *jeevamritha* samples and interacted with farmers. The natural farming and conventional farming plots were identified as told by the farmers and state officials. All the samples were analyzed in triplicate.

5.1.2 Protocol of measurements for plant nutrient concentration

Collected plant samples (for paddy, aboveground whole plant at 40-50 days after transplanting; for sugarcane, leaf samples of almost 120-130 days after planting; for sorghum, above ground biomass at harvest stage; and for turmeric, leaf samples of almost maturity stage, were oven-dried, finely ground and digested by wet digestion method. The contents of different plant nutrients were computed. Nitrogen concentration (%) in the plant samples was determined by micro-Kjeldahl distillation method after destroying the organic matter by H_2SO_4 and H_2O_2 (Piper, 1966). For the estimation of phosphorus, plant samples were digested with a di-acid mixture comprising of HNO_3: $HClO_4$ in the ratio of 3:1. Phosphorus concentration in the diluted di-acid digest was determined by developing yellow colour with Barton's reagent. The intensity of yellow colour was determined by using UV-VIS spectrophotometer at 420 nm (Piper, 1966) and expressed as percent. Potassium concentration in the diluted di-acid digest was determined by using flame photometer (Systronic make) and expressed as percent (Piper, 1966). Concentrations of micro-nutrient (Fe, Cu, Zn, Mn) in the diluted di-acid digest were determined by using Atomic Absorption Spectrophotometer following the method outlined by Lindsay and Norvell (1978). Boron in plant samples were determined by DTPA-Sorbitol extraction method (Miller *et al.*, 2001).

Annexure II. Protocol of measurements for soil chemical parameters

S. No.	Soil properties	Protocol	Reference
1.	Organic carbon	Wet digestion	Walkley and Black, 1934
2.	Available nitrogen	Kjeldahl method	Subbaiah and Asija, 1956
3.	Available phosphorus	0.5 M $NaHCO_3$ extraction	Olsen *et al.*, 1954
4.	Available potassium	Neutral normal ammonium extraction	Hanway and Heidal, 1952
5.	Exchangeable calcium	Ammonium acetate extraction	Lanyon and Heald, 1982
6.	Exchangeable magnesium	Ammonium acetate extraction	Lanyon and Heald, 1982
7.	Available sulphur	Turbidimetric method	Chesnin and Yien, 1950
8.	Available Boron	Azomethrine-H-hot water method	Berger and Truog, 1939
9.	DTPA-Zinc	DTPA-TEA-$CaCl_2$ extraction, AAS detection	Lindsay and Norvell, 1978
10.	DTPA-Iron	DTPA-TEA-$CaCl_2$ extraction, AAS detection	Lindsay and Norvell, 1978
11.	DTPA-Copper	DTPA-TEA-$CaCl_2$ extraction, AAS detection	Lindsay and Norvell, 1978
12.	DTPA-Manganese	DTPA-TEA-$CaCl_2$ extraction, AAS detection	Lindsay and Norvell, 1978

Annexure IVa. Plot-wise availability of soil organic carbon (SOC) and major nutrients in paddy cultivated fields in Vizianagaram district

S.No	Village code	SOC (%)		N (kg/ha)		P (kg/ha)		K (kg/ha)	
		NF	Non-NF	NF	Non-NF	NF	Non-NF	NF	Non-NF
1	GRL	0.67	0.78	194	220	35	44	171	248
2	PTD	0.40	0.61	207	232	31	39	123	141
3	PSRP	0.63	0.75	157	194	20	37	188	225
4	PDSL	0.50	0.68	207	220	18	23	142	229
5	KGP	0.65	0.71	169	194	23	29	155	191
6	SKR	0.50	0.76	169	207	24	32	161	225
7	GLD	0.70	0.87	220	244	33	38	175	226
8	AGR	0.81	0.71	298	283	33	28	254	213
9	KDP	0.74	0.88	260	294	29	35	255	306
10	PTR	0.44	0.48	248	263	18	22	162	173
11	GLN	0.67	0.78	194	220	35	44	171	248
	Range (n=11)	0.40-0.81	0.48-0.88	157-298	194-294	18-35	22-44	123-255	141-306
	Mean	0.61	0.73	211	234	27	34	178	220
	SD	0.13	0.11	43	34	7	8	42	43

Annexure IVb. Plot-wise availability of secondary nutrients in paddy cultivated fields in Vizianagaram district

S.No	Village Code	S (kg ha^{-1})		Ca (meq 100 g^{-1})		Mg (meq 100 g^{-1})	
		NF	Non-NF	NF	Non-NF	NF	Non-NF
1	GRL	16.1	17.8	19.6	20.4	5.0	6.1
2	PTD	12.4	11.9	17.8	16.6	4.4	4.2
3	PSRP	23.5	25.9	18.0	19.0	7.2	7.8
4	PDSL	19.2	24.1	12.2	11.4	5.2	4.9
5	KGP	17.8	17.1	17.4	18.8	6.6	7.2
6	SKR	22.4	23.7	11.0	12.2	5.2	6.6
7	GLD	12.8	15.7	5.8	6.2	2.2	2.6
8	AGR	27.5	29.1	6.8	7.9	2.2	2.5
9	KDP	32.5	31.6	6.5	8.2	2.3	2.6
10	PTR	23.6	25.9	10.5	11.3	3.8	4.8
11	GLN	18.9	20.8	16.6	18.4	5.0	6.3
Range (n=11)		12.4-32.5	11.9-31.6	5.8-19.6	6.2-20.4	2.2-7.2	2.5-7.8
	Mean	20.61	22.15	12.93	13.67	4.46	5.05
	SD	6.09	6.04	5.17	5.13	1.71	1.91

Annexure IVc. Plot-wise availability of soil micronutrients in paddy cultivated fields in Vizianagaram district

S. No	Village code	Fe (mg/kg)		Mn (mg/kg)		Cu (mg/kg)		Zn (mg/kg)		B (mg/kg)	
		NF	Non-NF	NF	Non-NF	NF	Non-NF	NF	Non-NF	NF	Non-NF
1	GRL	40.1	25.9	9.4	6.0	2.20	2.15	0.84	1.07	0.57	0.62
2	PTD	23.5	23.5	7.0	7.5	1.72	1.60	1.71	2.34	0.46	0.51
3	PSRP	26.0	23.3	8.3	6.1	1.51	1.73	1.10	2.19	0.63	0.66
4	PDSL	19.7	41.1	7.3	5.6	1.64	1.46	1.38	1.13	0.68	0.73
5	KGP	56.7	48.2	6.3	10.9	1.21	1.87	1.56	2.14	0.56	0.55
6	SKR	59.1	49.2	7.9	10.2	1.35	1.89	1.58	1.64	0.67	0.70
7	GLD	97.2	116.1	19.0	35.8	1.76	2.91	1.41	2.97	0.75	0.81
8	AGR	64.0	71.1	27.5	33.8	4.08	7.08	0.48	0.58	0.84	0.87
9	KDP	80.5	34.9	14.5	32.6	4.16	9.04	0.26	0.34	0.73	0.71
10	PTR	46.1	71.4	9.6	5.6	9.16	11.20	0.46	0.74	0.77	0.82
11	GLN	29.9	40.9	9.4	6.0	2.20	2.15	1.07	0.84	0.84	0.79
	Range (n=11)	19.7-97.2	23.3-116.1	6.3-27.5	5.6-35.8	1.21-9.16	1.46-11.20	0.26-1.71	0.34-2.97	0.46-0.84	0.51-0.87
	Mean	49.3	49.6	11.5	14.6	2.82	3.92	1.08	1.45	0.68	0.71
	SD	24.9	27.7	6.5	12.7	2.34	3.48	0.51	0.85	0.12	0.11

Annexure IVd. Plot-wise soil microbial population in paddy cultivated fields in Vizianagaram district

S. No.	Village code	Bacteria (Log_{10} CFU/g soil)		Fungi (Log_{10} CFU/g soil)		Actinomycetes (Log_{10} CFU/g soil)	
		NF	Non-NF	NF	Non-NF	NF	Non-NF
1	GRL	8.00	8.18	4.06	4.76	4.78	4.40
2	PTD	8.85	7.00	4.06	4.86	3.70	4.00
3	PSRP	9.04	8.54	4.02	4.34	4.85	4.48
4	PDSL	8.40	8.00	3.90	4.00	4.54	4.54
5	KGP	8.54	8.40	4.19	3.18	4.40	4.40
6	SKR	8.30	8.93	4.22	4.32	3.70	4.39
7	GLD	8.40	9.79	4.24	4.24	4.65	4.40
8	AGR	9.13	9.41	3.54	4.30	4.40	4.74
9	KDP	9.41	8.81	4.87	3.81	4.54	4.88
10	PTR	9.04	9.04	4.13	3.95	4.70	4.54
11	GLN	8.78	8.85	3.18	3.18	4.54	4.18
	Range (n=11)	8.00-9.41	7.00-9.79	3.18-4.87	3.18-4.86	3.70-4.85	4.00-4.88
	Mean	8.72	8.63	4.04	4.09	4.44	4.45
	SD	0.40	0.71	0.40	0.52	0.37	0.23